



CHANDU BAYA REDDY

(he/him)

Grazerstr. 1, Erlangen, 91052

Phone: +49 15566098124 **Email:** chandu.baya.reddy@gmail.com

[linkedin.com/in/chandu-b-reddy-a42291170/](https://www.linkedin.com/in/chandu-b-reddy-a42291170/) github.com/Chandubreddy819

About

M.Sc. ICT student (4th semester) at FAU Erlangen with expertise in communication systems, embedded development, and machine learning. Industry-proven skills from Bosch in designing embedded control units and optimising test systems for efficiency. Strong in Python, C/C++, MATLAB, and IoT hardware/software integration. Available for working student roles 20hrs/week during semester; expected graduation September 2026

Education

FAU Erlangen-Nuremberg

Master of Science in Information and Communication Technology

Major: Networks & Digital Communication

April 2024 – current

Erlangen, Bayern, Germany

Reva University, Bengaluru

Bachelor of Technology in Electronics and Communication

Major: Digital signal Processing, Wireless communications

August 2018 – June 2022

Bengaluru, Karnataka, India

GPA: 1.4

Current Projects

Joint Bluetooth & Mioty Localisation systems

July 2025 – Ongoing

Tech Stack: BLE, mioty, STM32, Python, C/C++, Ki Cad, Altium Designer, Leaflet.js, Web Visualization.

- Designing a Ultra-low power PCB for longer life cycles which includes battery management, LDO selection and sleep modes. Optimizing the Power management with Algorithms in Software stack other than its main application.
- Developing BLE + mioty localization tags (CR2032-powered) and algorithms, achieving sub-2m accuracy; visualized in real-time via Leaflet.js map

Audio signal processing - Speech synthesis and Analysis

April 2025 – July 2025

Tech Stack: Python, Matlab, Librosa, Matplotlib, scipy, Sound-Stream, LaTeX.

- Implemented STFT-based analysis, noise profiling, and objective metrics (SNR, SDR); automated figure/report generation (MATLAB) at LIKE Chair, AudioLabs.
- Presented a scientific paper based on Binaural Speech Synthesis.

Professional Experience

Robert Bosch

August 2022 - August 2023

Graduate Apprentice

Bengaluru, India

Product Development | Sensors | Power-Train Solutions

Technical skills: Raspberry Pi, Arduino (various other add-on boards), Python libs like NumPy, pandas, matplotlib, PyQt5 etc., CAN, Linux, Simulink, EasyEDA and JIRA.

- Designed & validated **sensor test control units**; built automated measurement routines (Python/MATLAB) to acquire and analyze signals, cutting test time 40% and cost 50%.
- Performed **electrical measurements** (CAN/SENT sensors), executed calibration of benches & probes, and documented **traceable procedures** and results
- Led failure analysis / **root-cause investigations** on noisy/unstable signals (clipping, drift, wiring faults) using spectral analysis, logging, and reproducible test scripts..
- Produced reports with plots, acceptance criteria, and equipment calibration records for release readiness.

Technical Skills

Measurements & Test: Acoustical/electrical measurements, calibration, equipment monitoring, data logging, failure analysis, root-cause, traceability.

Analysis & Tools: MATLAB (signal/acoustic analysis, bash scripts), Python (NumPy, SciPy, Matplotlib, Pandas), Simulink, Excel/PowerPoint reporting.

Hardware/Embedded: Microcontrollers (Arduino/RPi, STM32), CAN/SENT, I2C, SPI, UART, PCB Design, lab instrumentation usage.

Programming & Development: Python | C, C++ | Embedded C.

Machine Learning: TensorFlow, PyTorch, Machine Learning, Deep Learning.

Software & Tools: VS Code, PyCharm, EasyEDA, LTspice, Altium Designer, kiCAD, Linux (Ubuntu/Debian), Bash Scripting, Git/GitHub, LaTeX.

Data & Productivity Tools: Microsoft Office Suite (Word, Excel, PowerPoint), Power BI, JIRA.

Projects (Bachelors)

Solar Powered Wheelchair with Line Follower system with RF-ID Authentication Nov 2021 – April 2022

Tech Stack: Arduino-Uno, RC-522, IR sensor array, Motor-driver L298N, Relay 10A/5V, ESP32-cam, Solar panels.

- Developed autonomous navigation wheelchair powered by solar energy with RFID-based route authentication.
- Integrated line-following and obstacle avoidance system with live video streaming.

Hand Gesture controlled robot car with ESP32Cam module April 2020 – August 2020

Tech Stack: Arduino-nano, LilyPad Arduino, MPU6050 (Gyroscope & Accelerometer), Motor-drivers, Wi-Fi

- Implemented motion-based control via MPU6050 sensors, enabling intuitive navigation.
- Integrated live-streaming camera for remote control and monitoring.

Participations & Publications

- Participated in E-yantra Robotics Competition organized by **IIT Bombay**.
- Engaged in Smart-irrigation mini project focused on real-time analysis of IOT sensors, hosted by **Smart India Hackathon**.
- Published a Scientific paper on my Bachelor's major project "Solar Powered Wheelchair with Line Follower system with RFID Authentication" Article ID: IJERTCONV10IS11104. **IJERT**.

Language Skills

Mother Tongue: Kannada, Telugu

Other Languages: English - C1, German - A2, Hindi

Accomplishments

Awards: Employee of the Month (Robert Bosch) – for contributions to Embedded Systems & Sensor Testing in Powertrain Solutions.

Clean energy projects: Received Project funding for “Tertiary waste-water treatment automation” project from Karnataka State Council for Science and Technology (KSCST).